

Carotenoid Biosynthesis I — The Carotene Backbone (GGPP → Phytoene → Lycopene → Cyclic Carotenes)

A commissioned review-style synthesis for a molecular-biology audience

1. Executive Summary

The **carotene backbone** is the committed core of carotenoid biosynthesis: the segment that converts two molecules of the C₂₀ isoprenoid geranylgeranyl diphosphate (**GGPP**) into the C₄₀ cyclic carotenes (β-, α-, and γ-carotene) that seed every downstream carotenoid class. It comprises exactly three chemical operations executed in a fixed order: (i) a head-to-head condensation of two GGPP into 15-*cis*-phytoene by **phytoene synthase (PSY/CrtB)**, the first committed and rate-limiting step; (ii) four sequential desaturations, accompanied by *cis*→*trans* isomerizations, that convert colorless phytoene into red **all-*trans*-lycopene**; and (iii) cyclization of one or both ends of lycopene into β- and/or ε-ionone rings by **lycopene cyclases**. Everything upstream (GGPP supply from the MEP/MVA isoprenoid pools) and everything downstream (hydroxylation, epoxidation, ketolation, and oxidative cleavage to apocarotenoids such as retinal, abscisic acid, and strigolactones) lies outside the module.

The defining feature of the backbone — and the single most important organizing principle for readers — is that the pathway solves the same chemical problem by **two mutually exclusive but functionally interchangeable desaturation routes**. Most bacteria and fungi use a single, FAD-dependent, membrane-peripheral **CrtI-type desaturase** that performs all four dehydrogenations and delivers all-*trans*-lycopene directly. Cyanobacteria, algae, and plants instead use a **four-enzyme poly-*cis* route** — two plastoquinone-dependent desaturases (**PDS/CrtP** and **ZDS/CrtQ**) that each introduce two double bonds, plus two isomerases (**Z-ISO** and **CRTISO/CrtH**) that convert *cis* intermediates to the all-*trans* product. That the bacterial CrtI can substitute for the entire four-enzyme plant machinery *in planta* (the basis of Golden Rice) is direct proof that the two routes are functionally equivalent at the level of net chemistry, even though they differ completely in enzymology, cofactors, electron acceptors, and regulatory/signaling consequences.

Cyclization is likewise solved by **at least four structurally unrelated cyclase families** (CrtY; CrtL-type, including plant LCYB and LCYE; the fungal bifunctional phytoene-synthase/cyclase CrtYB/al-2; and the CruA/CruP family of green sulfur bacteria and cyanobacteria). The pathway has deep prokaryotic roots; the plant plastid machinery was inherited from cyanobacteria via endosymbiosis. Below we lay out the boundaries, mechanism, molecular players, evolutionary and cell-biological variation, ordering constraints, and open controversies, anchoring each major claim in primary literature.

2. Definition and Biological Boundaries

2.1 What is included

The carotene backbone module is defined by three obligatory transformations and the enzymes that carry them out:

Step	Substrate → Product	Enzyme family	Chemistry
Commitment	2 × GGPP → 15- <i>cis</i> -phytoene	PSY / CrtB	Head-to-head condensation
Desaturation	15- <i>cis</i> -phytoene → all- <i>trans</i> -lycopene	CrtI or (PDS + Z-ISO + ZDS + CRTISO)	4 dehydrogenations + isomerizations
Cyclization	lycopene → β-/α-/γ- carotene	CrtY / CrtL (LCYB, LCYE) / CrtYB / CruA- CruP	β- and/or ε-ring formation

The output of the module — the cyclic carotenes — is the branch point that feeds all downstream carotenoid chemistry.

2.2 What is adjacent but excluded

Several neighboring processes are frequently discussed alongside the backbone but are mechanistically separate and should be treated as boundaries, not contents:

- **GGPP supply.** The provision of GGPP from the plastidial MEP pathway (or cytosolic MVA in fungi) and by GGPP synthase is upstream. Notably, PSY does not act in isolation from this

supply: PSY–GGPPS protein–protein interaction strength is itself a determinant of flux (PMID: 40826491), which blurs the boundary at the commitment step but does not move it.

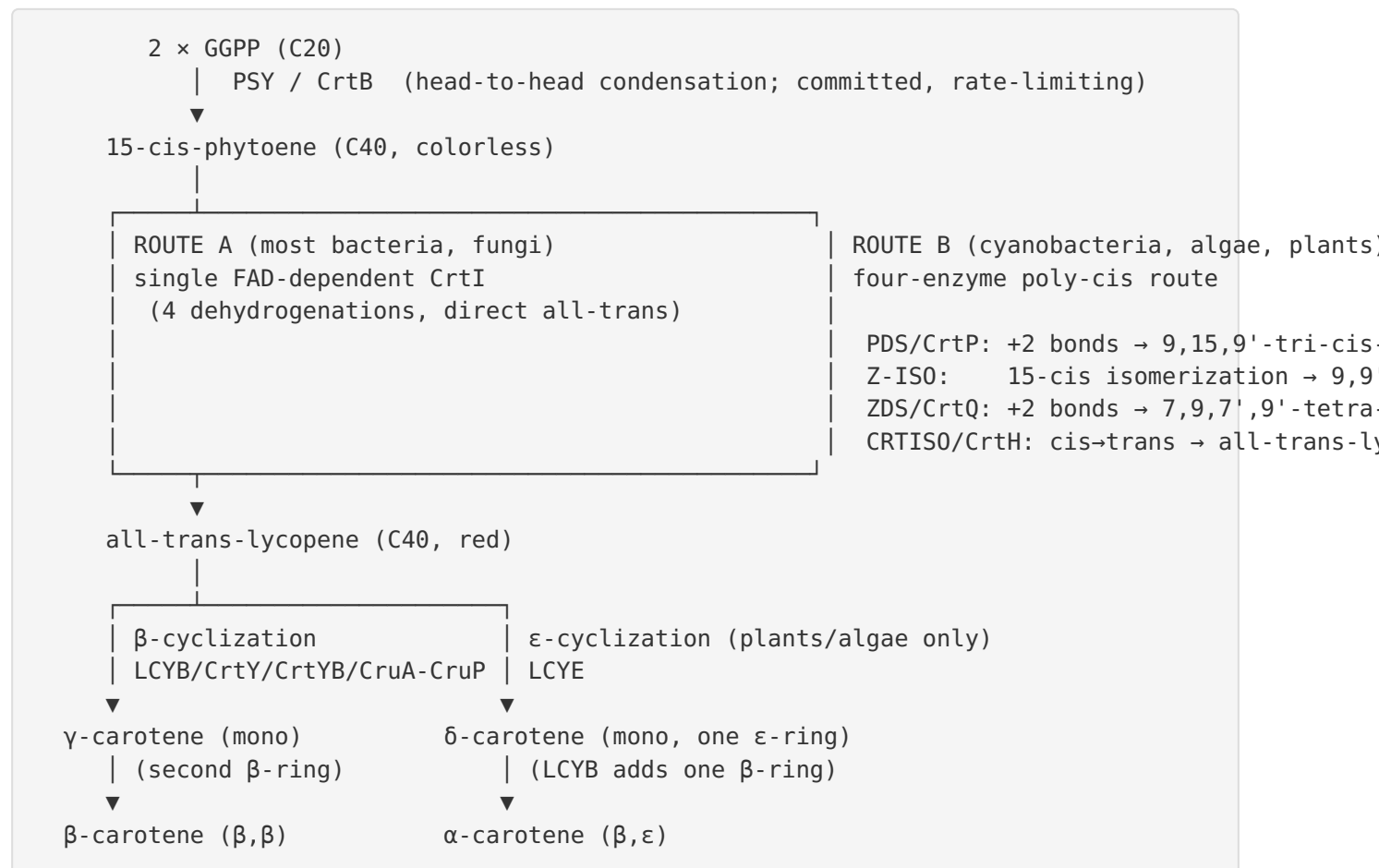
- **Xanthophyll formation.** Ring hydroxylation (non-heme di-iron CHY enzymes and P450 CYP97 hydroxylases), epoxidation (the xanthophyll cycle), and ketolation act on the cyclic carotenes and are downstream (PMID: 21450689, PMID: 22513258).
- **Apocarotenoid cleavage.** Oxidative cleavage by carotenoid cleavage dioxygenases to retinal, abscisic acid, strigolactones, and neurosporaxanthin is downstream (PMID: 33258195). The one caveat — important and returned to below — is that *linear cis-carotene intermediates of the desaturation step are themselves a substrate for a signaling apocarotenoid*, so the module has an internal regulatory output that is not merely feed-forward.
- **Photosynthetic function of carotenoids.** The structural/photoprotective roles of carotenoids in PSI/PSII reaction centers and light-harvesting complexes (PMID: 23896007) are consequences of the products, not part of the biosynthetic module.

2.3 Competing definitions

The literature is largely consistent on the boundaries, but there are two genuine ambiguities. First, whether **PTOX (plastid terminal oxidase)** belongs "inside" the module: it is not a carotenoid enzyme, yet it is an obligatory redox cofactor of the plant desaturation step (§6), so functionally it is part of the machinery even though it is physically a component of the plastoquinone/photosynthetic electron-transport system. Second, in fungi the **commitment and cyclization steps are fused** into a single bifunctional CrtYB/al-2 polypeptide (PMID: 11862485), so the "three separate operations" framing is a plant/bacterial abstraction rather than a universal architecture.

3. Mechanistic Overview

3.1 The best current model, step by step



3.2 Obligatory, conditional, and accessory steps

- **Obligatory:** PSY-catalyzed condensation (no phytoene, no pathway); the four net desaturations (colorless phytoene must become colored lycopene); at least one cyclization to make a cyclic carotene.
- **Conditional (route-dependent):** In Route B all four enzymes are individually required to reach all-*trans*-lycopene — but the *entire* Route B set is dispensable if Route A's CrtI is present, because CrtI performs the whole conversion by itself. This is the strongest single piece of evidence that the two routes are alternatives rather than complements.

- **Conditional (branch-dependent):** ϵ -cyclization is present only in lineages that make α -carotene/lutein (plants, green algae, some others). It is absent from the many bacteria and fungi that make only β -ring carotenoids.
- **Accessory / regulatory:** The **ORANGE (OR) chaperone**, which stabilizes PSY post-transcriptionally, and **PTOX**, which re-oxidizes the plastoquinone pool for the plant desaturases, are not catalytic backbone enzymes but are functionally required for normal flux in their respective systems.

3.3 Why the *cis* geometry matters

A crucial mechanistic subtlety is that **PSY produces 15-*cis*-phytoene, not the all-*trans* isomer**, and the plant desaturation route deliberately traffics in poly-*cis* intermediates that must be isomerized back to all-*trans*. The *cis* configuration is not incidental: Z-ISO and CRTISO exist precisely to correct geometries that the desaturases cannot. In Route A, CrtI can itself act as a *cis*→*trans* isomerase (anaerobically) and thus builds the isomerization function into the same polypeptide ([PMID: 22745782](#)).

4. Major Molecular Players and Active Assemblies

4.1 Phytoene synthase (PSY / CrtB): the committed gate

PSY condenses two GGPP molecules into 15-*cis*-phytoene, and this is universally described as the **first committed and rate-limiting step** of carotenogenesis ([PMID: 35498681](#)). Because it is rate-limiting, PSY output governs total carotenoid flux, and the enzyme is heavily regulated. A central mode of control is **post-transcriptional**: PSY levels and activity are fine-tuned without changing its localization to the plastid stroma and protothylakoid membranes, including through binding within a **PSY:ORANGE multi-enzyme complex** ([PMID: 37948577](#)). The OR protein acts as a holdase chaperone that stabilizes PSY and promotes chromoplast biogenesis ([PMID: 42165846](#)). In addition, PSY activity depends on productive interaction with GGPPs; in peach, differences in PSY1/PSY2 activity track the strength of the PSY–GGPPs interaction, and single interface residues can shift activity ([PMID: 40826491](#)). PSY is thus best understood not as a soluble solo enzyme but as a membrane-associated node embedded in protein–protein interaction networks.

4.2 The two desaturation routes

Route A — CrtI (single enzyme). The apo-CrtI crystal structure from *Pantoea ananatis* places CrtI in the **flavoprotein superfamily** together with protoporphyrinogen IX oxidoreductase and monoamine oxidase. It is a **membrane-peripheral oxidoreductase that uses FAD as its sole redox-active cofactor**, with O₂ (replaceable by quinones) as the terminal electron acceptor, and can act as a *cis-trans* isomerase under anaerobic conditions (PMID: 22745782). One enzyme therefore accomplishes both the four dehydrogenations and the geometric correction. An instructive variation: in *Myxococcus xanthus* two distinct CrtI-type desaturases cooperate — one acting on *cis*, one on *trans* substrates — to complete the four steps (PMID: 17662111), showing that even "the single-enzyme route" is not monolithic.

Route B — the four-enzyme poly-*cis* system. In cyanobacteria, algae, and plants, the conversion is partitioned:

Enzyme	Reaction	Notable mechanism
PDS / CrtP	15- <i>cis</i> -phytoene → 9,15,9'-tri- <i>cis</i> -ζ-carotene (+2 bonds)	Plastoquinone-dependent desaturase
Z-ISO	9,15,9'-tri- <i>cis</i> - → 9,9'-di- <i>cis</i> -ζ-carotene	Heme b-dependent isomerase of the 15-15' bond
ZDS / CrtQ	9,9'-di- <i>cis</i> -ζ-carotene → 7,9,7',9'-tetra- <i>cis</i> -lycopene (+2 bonds)	Plastoquinone-dependent desaturase
CRTISO / CrtH	prolycopene (tetra- <i>cis</i>) → all- <i>trans</i> -lycopene	Redox-type isomerase, structurally related to CrtI

Z-ISO is a *bona fide* integral-membrane enzyme that independently isomerizes the 15-15' *cis* double bond, and — remarkably for a carotenoid enzyme — catalysis depends on a **ferrous heme b cofactor with redox-regulated ligand switching** (PMID: 26075523). CRTISO, cloned as the tomato *tangerine* locus, is a redox-type enzyme structurally related to bacterial CrtI and is required to convert prolycopene to all-*trans*-lycopene; *tangerine* mutants accumulate prolycopene instead of lycopene (PMID: 11884678). Z-ISO is specifically required by oxygenic phototrophs and is absent from anoxygenic species such as green sulfur bacteria, which use CrtP/CrtQ/CrtH but not Z-ISO (PMID: 31593237, PMID: 36144332).

4.3 Lycopene cyclases: four unrelated solutions

Cyclization is the most striking case of **convergent enzymology** in the pathway. Four families of carotenoid cyclase are known, and *each family includes both mono- and dicyclases* that catalyze formation of γ - and β -carotene respectively (PMID: 18676669):

1. **CrtY** — bacterial (e.g., *Pantoea*) lycopene β -cyclase.
2. **CrtL-type** — including plant **LCYB** (β -cyclase) and **LCYE** (ϵ -cyclase). LCYB is the major β -cyclase in Arabidopsis (PMID: 19549928).
3. **Fungal bifunctional CrtYB / al-2** — a single polypeptide with both phytoene synthase and lycopene cyclase domains; mutations in the cyclase domain of *Neurospora* al-2 abolish cyclization and accumulate an acyclic carotenoid (PMID: 11862485).
4. **CruA / CruP** — members of the **FixC dehydrogenase superfamily**, only distantly related to CrtL/CrtY, that fill the cyclization gap in green sulfur bacteria and cyanobacteria (PMID: 17606904).

The **β/ϵ distinction** determines the downstream carotenoid class. A β -ring plus a β -ring gives β -carotene (β,β) and the β,β -xanthophylls (zeaxanthin, violaxanthin). One ϵ -ring plus one β -ring gives α -carotene (β,ϵ) and lutein. The **order is fixed in the α -branch**: LCYE forms one ϵ -ring first (giving δ -carotene), and LCYB then completes the molecule with a β -ring (giving α -carotene). Plant LCYE typically forms only a single ϵ -ring; interestingly, the liverwort *Marchantia polymorpha* LCYE can form two ϵ -rings (δ - $\rightarrow$ ϵ -carotene), unlike Arabidopsis LCYE (PMID: 24285752), showing lineage-specific variation in cyclase specificity.

5. Evolutionary and Cell-Biological Variation

5.1 Deep origin and inheritance

Phylogenetic analysis of carotenoid pathway genes indicates **early evolutionary roots in prokaryotes** (PMID: 34324713). With the rise of oxygenic photosynthesis in cyanobacteria and the subsequent endosymbiotic origin of plastids, the **cyanobacterial carotenoid pathway was inherited into algal and plant plastids** (PMID: 34324713). This explains why plant desaturation is cyanobacterial in type (PDS/ZDS/Z-ISO/CRTISO, plastoquinone-coupled) rather than fungal/bacterial (CrtI, FAD-coupled): the plant machinery is literally cyanobacterial machinery relocated to the plastid.

5.2 The route dichotomy as a lineage marker

Feature	Route A (CrtI)	Route B (poly- <i>cis</i>)
Distribution	Most bacteria, fungi	Cyanobacteria, algae, plants
# enzymes	1	4 (PDS, Z-ISO, ZDS, CRTISO)
Cofactor	FAD	Plastoquinone (desaturases); heme b (Z-ISO)
Electron sink	O ₂ / quinones	Plastoquinone → PTOX
Intermediates	mostly all- <i>trans</i>	poly- <i>cis</i>
Isomerase	built into CrtI	separate (Z-ISO, CRTISO)

The comparative enzymology across taxa is explicit: CrtP, CrtQ and isomerase CrtH form lycopene in most cyanobacteria, PDS/ZDS/Z-ISO/CrtISO in most algae and plants, while a single CrtI type is used in most bacteria and fungi ([PMID: 29741830](#)). Green sulfur bacteria are an informative intermediate: they are anoxygenic yet use the plant-like CrtP/CrtQ/CrtH set (but not Z-ISO), converting phytoene to lycopene via two plant-like desaturases and a plant-like isomerase — a pathway that "differs from the pathway known in all other bacteria" ([PMID: 15292122](#)).

5.3 Tissue, developmental, and compartmental variation

Within a plant, the same core enzymes operate in different plastid types and developmental contexts. PSY frequently exists as **tissue-specific isoforms** (e.g., PSY1/PSY2 in peach) with divergent activity driven by differential GGPPS interaction ([PMID: 40826491](#)). Cyclization branch usage is developmentally and tissue-tuned (leaf chloroplasts favor a lutein/ β , β balance; ripening fruit chromoplasts often accumulate lycopene or β -carotene). A key light/dark difference: because the plant route generates *cis* intermediates that can also isomerize photochemically, CRTISO is especially important for carotenoid synthesis **in the dark and in non-photosynthetic tissue**, where light-driven isomerization is unavailable ([PMID: 11884678](#)). Consistently, Z-ISO is required for **light-independent** carotenoid biosynthesis in cyanobacteria ([PMID: 36144332](#)).

5.4 The signaling dimension

A distinctive feature of Route B is that its **linear *cis*-carotene intermediates are themselves a source of a plastid-signaling apocarotenoid**. In Arabidopsis, disruption of **CRTISO** (but not Z-ISO) reduces chlorophyll in young leaves; the *ccr2* (CRTISO) phenotype is rescued by lowering PSY activity, implicating a specific acyclic *cis*-carotene-derived signal that modulates the **PIF3/HY5 module and plastid biogenesis** (PMID: 35177117, PMID: 37948577). This distinguishes CRTISO from Z-ISO mechanistically at the *organismal* level, not just the enzymatic one, and gives Route B a regulatory output that Route A (which never accumulates these intermediates) cannot produce (PMID: 33258195).

6. Constraints, Dependencies, and Failure Modes

6.1 Ordering constraints

- **Condensation before desaturation before cyclization.** Phytoene is the obligatory first C40; lycopene is the obligatory cyclase substrate. Cyclases act on the acyclic all-*trans*-lycopene (or on partially desaturated substrates in some bacteria), never on phytoene.
- **Within Route B, the order is strictly PDS → Z-ISO → ZDS → CRTISO.** Each enzyme's product is the next enzyme's substrate: PDS supplies tri-*cis*-ζ-carotene to Z-ISO; Z-ISO supplies di-*cis*-ζ-carotene to ZDS; ZDS supplies prolycopene to CRTISO. Skipping Z-ISO leaves the 15-15' *cis* bond uncorrected, and the route stalls in the dark.
- **In the α-branch, LCYE acts before LCYB.** The ε-ring is formed first (δ-carotene), then a β-ring is added (α-carotene). Two β-rings give β-carotene; the branch is decided by whether LCYE gets access to lycopene.

6.2 Compartment- and cofactor-specific dependencies

The plant desaturases are **redox-coupled to the photosynthetic electron-transport chain**. PDS and ZDS pass electrons to **plastoquinone**, which must be re-oxidized by the **plastid terminal oxidase (PTOX)**. The tomato *ghost* (*gh*) PTOX-null mutant is carotenoid-deficient with bleached fruit, consistent with PTOX acting as a phytoene-desaturase cofactor (PMID: 17873087). This coupling has a striking consequence for metabolic engineering: when bacterial CrtI is expressed in Arabidopsis, it too dumps electrons into the plastoquinone pool via PTOX,

and PTOX competes efficiently with cyclic electron flow for plastoquinol, raising reactive oxygen species and causing photoinhibition (PMID: 24378845). Thus even a "foreign" desaturase is forced into the host's redox constraints.

6.3 Mutual exclusivity and interchangeability — the central paradox resolved

The two routes are **mutually exclusive within a lineage** (organisms use one or the other) yet **functionally interchangeable across lineages**. The decisive experiment is Golden Rice: co-expressing plant PSY with bacterial **CrtI alone** was sufficient to drive β -carotene synthesis *and* downstream xanthophylls in rice endosperm, without any of the plant PDS/Z-ISO/ZDS/CRTISO enzymes (PMID: 11880581). The same two-gene (CrtB + CrtI) strategy enriches provitamin A in wheat (PMID: 24692648). This rules out the otherwise plausible idea that the plant four-enzyme system is *chemically necessary* in a plastid; it is a lineage-inherited implementation, not a requirement.

6.4 Failure modes

- **Loss of PSY** → no carotenoids at all (complete block at commitment).
- **Loss of any Route B enzyme** → accumulation of the corresponding *cis* intermediate (e.g., prolycopene in CRTISO-null *tangerine*; ζ -carotene isomers in Z-ISO-null), photobleaching, and — in CRTISO loss — aberrant *cis*-carotene signaling that lowers chlorophyll.
- **Loss of PTOX** → desaturation stalls for lack of an electron sink; bleached, carotenoid-deficient tissue (tomato *ghost*).
- **Misregulated flux** (e.g., unstable PSY without OR) → reduced carotenoid storage and chromoplast defects.

7. Mechanistic Model — Integrated Narrative

The carotene backbone is best understood as **one conserved chemical logic implemented by two enzymatic dialects**. The logic is invariant: commit isoprenoid flux (PSY), extend the conjugated polyene to lycopene by four desaturations with geometric correction, then close rings. The dialects differ in how the desaturation problem is solved. The **CrtI dialect** is compact and self-contained — a single FAD flavoprotein that dehydrogenates and isomerizes, dumping electrons onto O₂/quinones. The **poly-*cis* dialect** is distributed — two plastoquinone-

coupled desaturases wired into photosynthetic redox via PTOX, plus a heme-b isomerase (Z-ISO) and a CrtI-related isomerase (CRTISO) that repair the *cis* geometries the desaturases generate.

The two dialects are not chemically superior to one another — Golden Rice proves substitutability — but they carry different *side effects*. The poly-*cis* dialect couples carotenoid synthesis to the plastoquinone pool and generates *cis*-carotene intermediates that double as retrograde signals. These side effects, not the core chemistry, are probably what natural selection acts on when a lineage "keeps" a route. Cyclization then layers a second, independent case of convergence: four unrelated protein families all learned to fold lycopene's ends into β - or ϵ -rings, with the β/ϵ choice and its order defining the entire downstream carotenoid landscape.

8. Evidence Base — Key Literature

PMID	How it supports the review
29741830	Contrasts the single CrtI route with the four-enzyme poly- <i>cis</i> route across taxa (route dichotomy)
26075523	Z-ISO substrate/regiochemistry and heme b cofactor dependence
22745782	CrtI crystal structure: flavoprotein superfamily, FAD, membrane-peripheral, isomerase activity
17606904	CruA/CruP as a fourth, FixC-superfamily cyclase family
18676669	Four cyclase families; mono-/di-cyclase = γ/β distinction
35498681	PSY as first committed, rate-limiting step
37948577	Post-transcriptional PSY control and PSY:ORANGE complex; <i>cis</i> -carotene signal to PIF3/HY5
42165846	ORANGE holdase chaperone stabilizes PSY, promotes chromoplasts
40826491	PSY–GGPPS interaction strength drives isoform activity
34324713	Prokaryotic roots; endosymbiotic inheritance into plastids
17873087	PTOX as phytoene-desaturase cofactor (tomato <i>ghost</i>)
24378845	CrtI feeds electrons into plastoquinone pool via PTOX; ROS/photoinhibition
11880581	CrtI substitutes for whole plant desaturation/isomerization machinery (Golden Rice)
24692648	CrtB + CrtI two-gene strategy enriches provitamin A in wheat
35177117	CRTISO (not Z-ISO) loss lowers chlorophyll; <i>cis</i> -carotene retrograde signaling
11884678	CRTISO cloned as tomato <i>tangerine</i> ; polycopene accumulation; dark/non-photosynthetic role
31593237	Z-ISO required by oxygenic phototrophs, absent in anoxygenic species
36144332	Z-ISO required for light-independent carotenoid synthesis in cyanobacteria
15292122	Green sulfur bacteria use plant-like CrtP/CrtQ/CrtH (but not Z-ISO)
11862485	Fungal bifunctional PSY/cyclase (al-2); cyclase-domain mutants
17662111	Two cooperating CrtI-type desaturases in <i>Myxococcus</i>

PMID	How it supports the review
24285752	Liverwort LCYE forms two ϵ -rings (lineage-specific cyclase variation)
19549928	LCYB as major β -cyclase in Arabidopsis
18788479	Cyanobacterium cyclizes lycopene without a classical cyclase gene

9. Controversies and Open Questions

- 1. Where exactly is the module boundary at the commitment step?** Because PSY activity is governed by GGPPS interaction and by the OR chaperone, the "first committed step" is really a regulated multi-protein node. How much of carotenoid flux control lies in PSY catalysis versus PSY stabilization/complex assembly remains quantitatively unresolved ([PMID: 37948577](#), [PMID: 40826491](#)).
- 2. The identity and generality of the *cis*-carotene signal.** The acyclic *cis*-carotene-derived apocarotenoid that regulates the PIF3/HY5 module is functionally inferred but not chemically identified in most reports; whether a single molecule or a family of signals is responsible, and how broadly it operates beyond Arabidopsis, is open ([PMID: 35177117](#), [PMID: 33258195](#)).
- 3. Cyclase family boundaries and convergence.** With four unrelated cyclase families (including CruA/CruP in the FixC dehydrogenase superfamily), the ancestral cyclase and the number of independent inventions of ring formation are debated ([PMID: 17606904](#), [PMID: 18676669](#)). Some cyanobacterial genomes lack any classical cyclase gene yet still cyclize, indicating additional undiscovered players ([PMID: 18788479](#)).
- 4. Mixing organisms.** Much mechanistic detail comes from a few models (Arabidopsis, tomato, maize, *Pantoea*, *Synechocystis*, *Neurospora*). Z-ISO has been functionally characterized in only a handful of species ([PMID: 31593237](#)), and inferring universal mechanism from these is a real risk. The presence of plant-like desaturases in *anoxygenic* green sulfur bacteria — but without Z-ISO — warns against equating "plant-type route" with "oxygenic" in every detail ([PMID: 15292122](#), [PMID: 31593237](#)).

5. **Why keep the complicated route at all?** If CrtI can do the whole job, why did the cyanobacterial/plant lineage retain a four-enzyme, poly-*cis*, PTOX-dependent system? The signaling output of *cis* intermediates and integration with photosynthetic redox poise are candidate answers, but a rigorous adaptive explanation is not settled.

10. Limitations and Knowledge Gaps

- **Organismal narrowness.** The mechanistic backbone is assembled from a small set of model systems; several enzymes (notably Z-ISO) are characterized in only a few species, and cross-organism generalization should be made cautiously.
 - **This review is a literature synthesis, not a data analysis.** No new experimental or sequence data were generated; conclusions rest on published primary literature and reviews.
 - **Chemical identity of the *cis*-carotene signal** is unresolved, as is the quantitative apportioning of flux control between PSY catalysis and PSY stabilization.
 - **Cyclase completeness.** At least one cyanobacterium cyclizes lycopene without a recognizable classical cyclase, implying undiscovered enzymes ([PMID: 18788479](#)).
 - **Structural data are uneven.** High-resolution structures exist for CrtI but are sparse for the plant desaturases and most cyclases, limiting mechanistic modeling.
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11. Proposed Follow-up Experiments / Actions

1. **Chemically identify the acyclic *cis*-carotene retrograde signal** using targeted metabolomics in *ccr2*/CRTISO and Z-ISO backgrounds, and test candidate apocarotenoids for PIF3/HY5 modulation.
2. **Structural determination of PDS, ZDS, Z-ISO, and CRTISO** (cryo-EM of membrane-embedded forms) to explain plastoquinone binding and Z-ISO's heme-b redox switching.
3. **Systematic route-swap panel** — express CrtI vs. the full poly-*cis* set in a common plastid host and quantify redox load, ROS, and photoinhibition to test whether the poly-*cis* route's persistence reflects redox integration rather than chemistry ([PMID: 24378845](#)).
4. **Search cyanobacterial genomes lacking classical cyclases** for the missing cyclase activity implied by [PMID: 18788479](#).

5. **Quantitative flux-control analysis at the commitment node**, dissecting PSY catalysis, PSY–GGPPS interaction, and OR-mediated stabilization contributions to total carotenoid flux.
6. **Broaden Z-ISO characterization** across additional oxygenic and anoxygenic lineages to firm up the "oxygenic phototrophs need Z-ISO" generalization.

Prepared as a commissioned review synthesis. Claims are anchored to the cited primary literature; uncertainty is flagged where mechanistic detail derives from a single organism or indirect evidence.

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